

Vyra Platform Architecture v7

Architecture Narrative Baseline

Purpose

This document defines the architecture of Vyra, an Agentic Risk & Compliance Infrastructure platform.

It is not a graph schema specification, a product requirements document, or an implementation guide.

Its purpose is to explain how Vyra transforms compliance from a periodic, human-driven activity into a continuously operating autonomous system.

Core Thesis

Traditional compliance systems treat compliance as documentation.

Vyra treats compliance as an operational system.

Traditional Model:

Regulations → Humans → Spreadsheets → Audits → Reports

Vyra Model:

Regulations → Agents → Continuous Compliance → Assurance

The Throughline

The entire architecture is organized around a single continuous loop:

Regulations ↓ Requirements ↓ Execution ↓ Operations ↓ Signals ↓ Reasoning ↓ Risk ↓ Remediation
↓ Assurance ↓ Regulations

This is the Continuous Compliance Loop.

Every capability, service, graph, workflow, and agent within Vyra exists to support one part of this loop.

Architectural Principle

The platform should be explained through the progression:

Regulation → Execution → Observation → Reasoning → Action → Assurance

Rather than through technical components.

The reader should first understand:

1. Why compliance exists
2. How compliance work is performed
3. How operational reality is observed
4. How the platform reasons
5. How agents collaborate
6. How actions are executed
7. How assurance is produced

Only then should implementation details such as graph schemas, ontologies, Neo4j structures, and data models be introduced.

Role of the Graph

The graph is not the product.

The graph is the operating system of the platform.

The graph acts as:

- Enterprise Memory
- Compliance Memory
- Risk Memory
- Agent Collaboration Layer

Agents do not primarily communicate with one another.

Agents collaborate through the graph.

The graph is the shared source of truth.

Four Interconnected Graphs

Vyra is composed of four interconnected graph domains.

Regulatory Graph

Defines obligations.

Examples:

- Frameworks
 - Regulations
 - Requirements
 - Controls
-

Execution Graph

Defines compliance work.

Examples:

- Programs
 - Workflows
 - Tasks
 - Schedules
-

Operational Graph

Defines reality.

Examples:

- Assets
 - Signals
 - Evidence
 - Risks
-

Intelligence Graph

Defines reasoning.

Examples:

- Agents
 - Findings
 - Decisions
 - Remediations
-

Together these form the Unified Vyra Graph.

Agent Collaboration Model

Compliance intelligence does not emerge from a single agent.

It emerges through collaboration.

Agents contribute observations, findings, evidence, decisions, and remediations into the shared graph.

The graph becomes the collaboration medium.

This is a fundamental architectural differentiator of Vyra.

Document Objective

The objective of Vyra Platform Architecture v7 is to explain:

- What Vyra is
- Why it exists
- How it works
- How agents collaborate
- How the graph enables reasoning
- How compliance becomes autonomous

The document should tell a coherent story rather than present a collection of disconnected technical concepts.

Every chapter should advance the reader through the Continuous Compliance Loop.